自动驾驶决策规划算法第一章第三节（下）

$$l = (\vec{r}_h - \vec{r}_r)\cdot\vec{n}_r$$

$$\dot{l} = \vec{v}_h\cdot\vec{n}_r$$

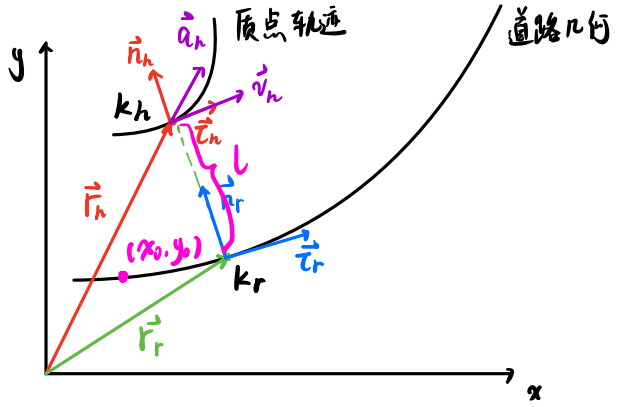
$$\dot{s} = \frac{\vec{v}_h\cdot\vec{\tau}_r}{1-k_r l}$$

$$\ddot{l} = \vec{a}_h\cdot\vec{n}_r - k_r(1-k_r l)\dot{s}^2$$

$$l' = \frac{\dot{l}}{\dot{s}}$$

$$\ddot{s} = \frac{\vec{a}_h\cdot\vec{\tau}_r}{1-k_r l} + \frac{\dot{s}^2 k_r l'}{1-k_r l} + \frac{\dot{s}^2(k_r' l + k_r l')}{1-k_r l}$$

$$l'' = \frac{\ddot{l} - l'\ddot{s}}{\dot{s}^2}$$



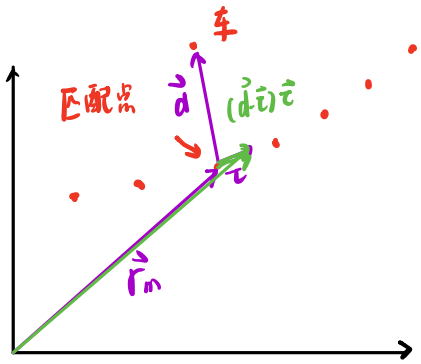
一、如何找到 x_r, y_r, θ_r, k_r ✓

二、如何计算 s ✓

三、如何从 $s, \dot{s}, \ddot{s}, l, \dot{l}, \ddot{l}, l', l''$ → $\vec{r}_h, \vec{v}_h, \vec{a}_h, k_h$

如何找到 x_r, y_r, θ_r, k_r，在《自动驾驶控制算法第七讲》有讲过

一般 reference line 都是离散点，只讲离散点的 x_r, y_r, θ_r, k_r



k_i 为曲率（直角坐标下）

θ_i 为切线方向与 x 轴的夹角

找距离最短的，此点称为匹配点 $(x_m, y_m, \theta_m, k_m)$

计算 $\vec{d}$，以及匹配点的切向量 $\vec{\tau}$

设车 (x_h, y_h) 则 $\vec{d} = (x_h - x_m, y_h - y_m)$

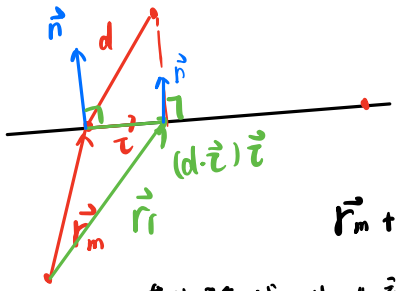
$$\vec{\tau} = (\cos\theta_m, \sin\theta_m)$$

则近似 $\vec{r}_r = \begin{pmatrix} x_r \\ y_r \end{pmatrix} = \vec{r}_m + (\vec{d}\cdot\vec{\tau})\vec{\tau}$

$$= \begin{pmatrix} x_m \\ y_m \end{pmatrix} + (\vec{d}\cdot\vec{\tau})\begin{pmatrix} \cos\theta_m \\ \sin\theta_m \end{pmatrix}$$

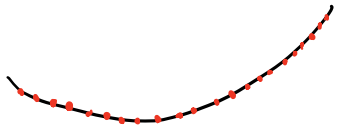
用 $\vec{r}_m + \vec{d}$ 在 $\vec{\tau}$ 上的向量分量近似代替 $\vec{r}_r$

原理



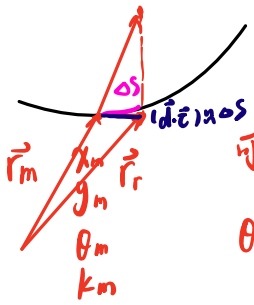
$\vec{r}_m + (\vec{d}\cdot\vec{\tau})\vec{\tau} = \vec{r}_f$ 在 frenet path 为直线时为精确解

若为弧线，当曲率不太大，且离散的点较密时，精度也较高



曲率越小，划分越密，每段越像直线，精度越高

求 θ_r, k_r



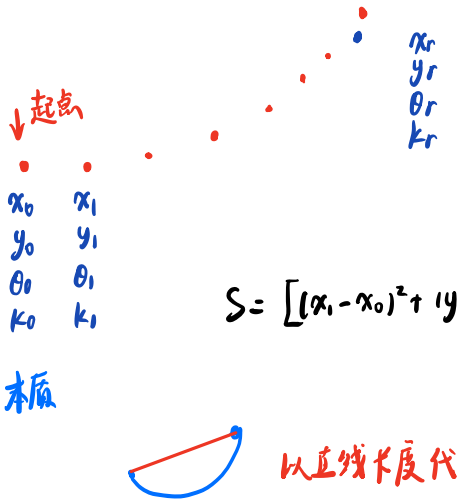
可近似认为 $k_r = k_m$

$\theta_r \approx \theta_m + k_m\cdot(\vec{d}\cdot\vec{\tau})$

$\because k_m = \frac{d\theta}{ds} \approx \frac{\Delta\theta}{\Delta s} \approx \frac{\Delta\theta}{(\vec{d}\cdot\vec{\tau})}$ $\therefore \Delta\theta = k_m\cdot(\vec{d}\cdot\vec{\tau})$

$\theta_r = \theta_m + \Delta\theta = \theta_m + k_m\cdot(\vec{d}\cdot\vec{\tau})$

如何计算 s



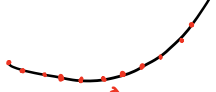
$S = [(x_1-x_0)^2 + (y_1-y_0)^2]^{\frac{1}{2}} + [(x_2-x_1)^2 + (y_2-y_1)^2]^{\frac{1}{2}} + \cdots$

本质

以直线长度代替弧长

当路径点足够密，误差可以忽略

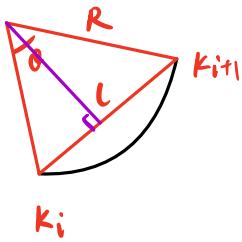




近似是一条直线

有其它更精确的算法，但是不稳定，以分段直线近似曲线虽然比较糙，但是对参数变化不敏感 (robust)

精确算法例子



$\bar{k} = \frac{k_i + k_{in}}{2}$ $R = \frac{1}{\bar{k}}$

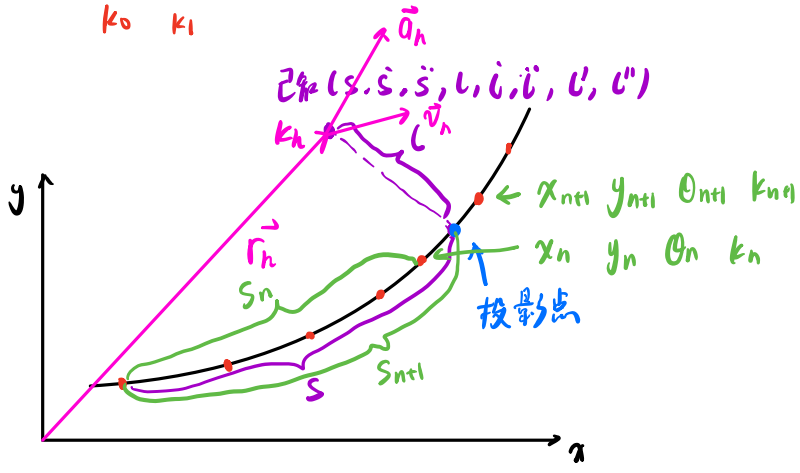
$l = 2R\sin\frac{\theta}{2}$ $\theta = 2\arcsin\frac{l}{2R}$

$s = R\theta$ 以圆弧近似曲线

问题 $\bar{k}$ 一般较小 $R = \frac{1}{\bar{k}}$ $\bar{k}$ 微小变化会引起 R 的剧烈波动，算法不稳定

已知 $s, \dot{s}, \ddot{s}, l, \dot{l}, \ddot{l}, l', l''$ } 求 $\vec{r}_n, \vec{v}_n, \vec{a}_n, \vec{k}_n$

已知 x_0 x_1 ...
y_0 y_1
θ_0 θ_1
k_0 k_1



Frenet 转 Cartesian

关键是还是找投影点的 x_r, y_r, θ_r, k_r

1. 计算 $x_i \to s_i$ 的对应表，即每个 frenet 坐标轴上离散点的 x_i 与 s_i 的对应关系

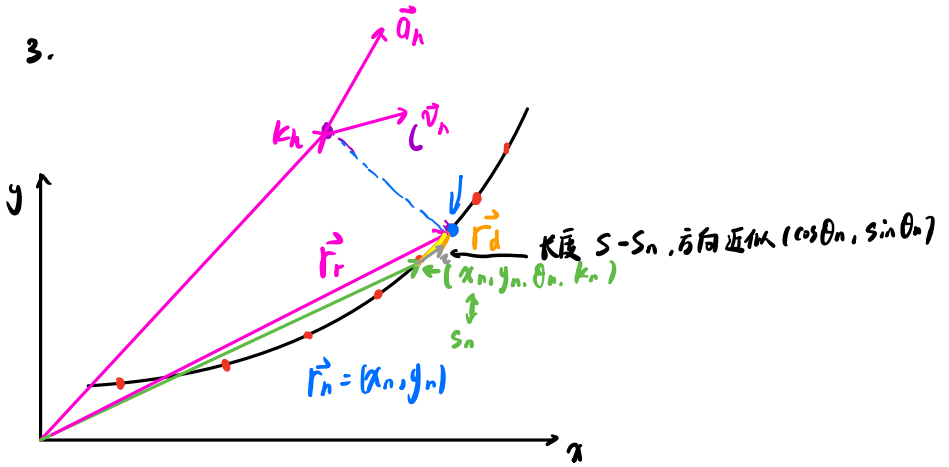
x_0	x_1	x_L
0	$[(x_1-x_0)^2+(y_1-y_0)^2]^{\frac{1}{2}}$	$\sum_{i=1}^{L}[(x_i-x_{i-1})^2+(y_i-y_{i-1})^2]^{\frac{1}{2}}$

2. 查找 $x_i \leftrightarrow s_i$ 表，找到 s_n，使得 $s_n < s < s_{n+1}$，s_n 与 s_{n+1} 对应的 x_i 记为 x_n, x_{n+1}

找到 x_n, y_n, θ_n, k_n

$x_{n+1}, y_{n+1}, \theta_{n+1}, k_{n+1}$

3.



$\vec{r}_d = (s-s_n)\vec{\tau}_n$ 式 $\bar{\theta} = \frac{\theta_n + \theta_{n+1}}{2}$ $\vec{\tau}_n = (\cos\bar{\theta}, \sin\bar{\theta})$

$\vec{r}_r = \vec{r}_d + \vec{r}_n$ 求出 x_r, y_r

$\theta_r = \theta_n + k_n \cdot (s-s_n)$

$k_r = \frac{k_n + k_{n+1}}{2}$

$k_r' = \frac{dk_r}{ds} \approx \frac{k_{n+1}-k_n}{s_{n+1}-s_n}$

$\vec{\tau}_r = (\cos\theta_r, \sin\theta_r)$ $\vec{n}_r = (-\sin\theta_r, \cos\theta_r)$

得到 $\vec{r}_r, \vec{\tau}_r, \vec{n}_r, k_r, k_r'$

4

$l = (\vec{r}_h - \vec{r}_r)\cdot\vec{n}_r$ -①

$\dot{l} = \vec{v}_h \cdot \vec{n}_r$ -②

$\dot{s} = \frac{\vec{v}_h \cdot \vec{\tau}_r}{1-k_r l}$ -③

$\ddot{l} = \vec{a}_h \cdot \vec{n}_r - k_r(1-k_r l)\dot{s}^2$ -④

$l' = \frac{\dot{l}}{\dot{s}}$ -⑤

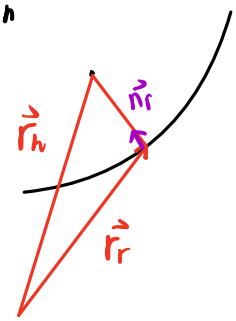
$\ddot{s} = \frac{\vec{a}_h \cdot \vec{\tau}_r}{1-k_r l} + \frac{\dot{s}^2 k_r l'}{1-k_r l} + \frac{\dot{s}^2(k_r' l + k_r l')}{1-k_r l}$ -⑥

$l'' = \frac{\ddot{l} - l'\ddot{s}}{\dot{s}^2}$ -⑦

已知 $s, \dot{s}, \ddot{s}, l, \dot{l}, \ddot{l}, l', l'', \vec{r}_r, \vec{\tau}_r, \vec{n}_r, k_r, k_r'$，求

$\vec{r}_h$, $\vec{v}_h$, $\vec{a}_h$, k_h

(1) 算 $\vec{r}_h$



$\vec{r}_h = \vec{r}_r + l\vec{n}_r$

(2) 算 $\vec{v}_h$

由 ② ③

$\dot{l} = \vec{v}_h \cdot \vec{n}_r \Rightarrow \vec{v}_h \cdot \vec{n}_r = \dot{l} \Rightarrow |\vec{v}_h||\vec{n}_r|\cos\langle \vec{v}_h, \vec{n}_r\rangle = \dot{l}$

$\dot{s} = \frac{\vec{v}_h \cdot \vec{\tau}_r}{1-k_r l} \quad \vec{v}_h \cdot \vec{\tau}_r = (1-k_r l)\dot{s} \Rightarrow |\vec{v}_h||\vec{\tau}_r|\cos\langle \vec{v}_h \cdot \vec{\tau}_r\rangle = (1-k_r l)\dot{s}$

设 $\vec{v}_h$ 与 x 轴夹角为 θ_h

∵ $\vec{\tau}_r$ 与 x 轴夹角 θ_r　$\vec{n}_r$ 与 $\vec{\tau}_r$ 垂直　∴ $\vec{n}_r$ 与 x 轴夹角 $\theta_r + \frac{\pi}{2}$

∴ $\cos\langle \vec{v}_h, \vec{n}_r\rangle = \cos\left(\theta_h - (\theta_r + \frac{\pi}{2})\right) = \sin(\theta_h - \theta_r)$

$\cos\langle \vec{v}_h, \vec{\tau}_r\rangle = \cos(\theta_h - \theta_r)$

带入

$$\begin{cases} |\vec{v}|\sin(\theta_h-\theta_r) = \dot{l} \\ |\vec{v}|\cos(\theta_h-\theta_r) = (1-k_r l)\dot{s} \end{cases}$$ 平方相加 $|\vec{v}| = [\dot{l}^2 + (1-k_r l)^2\dot{s}^2]^{\frac{1}{2}}$ 大小

再算方向　相除　$\tan(\theta_h-\theta_r) = \frac{\dot{l}}{(1-k_r l)\dot{s}} = \frac{l'}{1-k_r l}$

$\theta_h = \theta_r + \arctan\left(\frac{l'}{1-k_r l}\right)$

∴ $\vec{\tau}_h$ 方向 $= (\cos\theta_h, \sin\theta_h)$　$\vec{n}_h = (-\sin\theta_h, \cos\theta_h)$

(3) 算 $\vec{a}_h$

$\ddot{l} = \vec{a}_h \cdot \vec{n}_r - k_r(1-k_r l)\dot{s}^2$

$\ddot{s} = \frac{\vec{a}_h \cdot \vec{\tau}_r}{1-k_r l} + \frac{\dot{s}^2 k_r l'}{1-k_r l} + \frac{\dot{s}^2(k_r' l + k_r l')}{1-k_r l}$

$\vec{a}_h \cdot \vec{n}_r = \ddot{l} + k_r(1-k_r l)\dot{s}^2$

$\vec{a}_h \cdot \vec{\tau}_r = \ddot{s}(1-k_r l) - \dot{s}^2 k_r l' + \dot{s}^2(k_r' l + k_r l')$

同理 $|\vec{a}_n| = \{[\ddot{l} + k_r(1-k_r l)\dot{s}^2]^2 + [\ddot{s}(1-k_r l) - \dot{s}^2 k_r' l + \dot{s}^2(k_r' l + k_r l')]^2\}^{\frac{1}{2}}$

设 $\vec{a}_n$ 与 x 轴夹角为 θ_{an}

则 $\theta_{an} = \arctan\left(\frac{\ddot{l} + k_r(1-k_r l)\dot{s}^2}{\ddot{s}(1-k_r l) - \dot{s}^2 k_r' l + \dot{s}^2(k_r' l + k_r l')}\right) + \theta_r$

注意 博客中的 a_x，实际上是 $|\vec{v}|$

---- v_x，实际上是 $|\vec{v}|$

$|\vec{v}| \neq |\vec{a}_n|$ $\quad \vec{a}_n = \underbrace{|\vec{v}|\vec{\tau}_r}_{\text{切向}} + \underbrace{|\vec{v}|^2 k_n \vec{n}_r}_{\text{向心加速度}}$

计算出了 θ_{an} 也就计算出了 $\vec{\tau}_{an} = (\cos(\theta_{an}), \sin(\theta_{an}))$

$\vec{a}_n = |\vec{a}_n|\vec{\tau}_{an}$

(4) 算 k_n

$\vec{a} = |\vec{v}|\vec{\tau}_r + |\vec{v}|^2 k_n \vec{n}_r$

两边点乘 $\vec{n}_r$ $\quad \vec{a}\cdot\vec{n}_r = |\vec{v}|^2 k_n$ $\quad k_n = \frac{\vec{a}\cdot\vec{n}_r}{|\vec{v}|^2}$